Group Discussion Assignment

Evaluation Methods for Large Language Models (LLMs)

Number of Groups: 10

Topic: LLM Performance Evaluation

---

1. Introduction

Large Language Models (LLMs), such as GPT, Claude, Gemini, Llama, and Qwen, are widely used in natural language processing, text generation, machine translation, question answering, and many other applications.

However, evaluating the performance of these models requires appropriate evaluation methods and metrics.

In this assignment, students will research, understand, and implement different evaluation methods used to measure the performance of Large Language Models.

2. Topics for 10 Groups

Each group is assigned one research topic. Students must investigate the theoretical concepts, mathematical formulas, practical applications, and Python implementation of their assigned topic.

Group	Research Topic	Main Focus
Group 1	Exact Match (EM) Evaluation	Measuring exact answer matching between predictions and reference answers.
Group 2	Accuracy Evaluation	Measuring the proportion of correct predictions.
Group 3	Precision, Recall, and F1-score	Evaluating classification performance and balancing precision and recall.
Group 4	BLEU Score	Evaluating generated text using n-gram precision, especially in machine translation.
Group 5	ROUGE Score	Evaluating text summarization using word and sequence overlap.

Group	Research Topic	Main Focus
Group 6	METEOR Score	Evaluating text generation using word matching, stemming, synonyms, and word order.
Group 7	BERTScore	Evaluating semantic similarity using contextual language model embeddings.
Group 8	Word Error Rate (WER)	Measuring word-level insertion, deletion, and substitution errors.
Group 9	Character Error Rate (CER)	Measuring character-level insertion, deletion, and substitution errors.
Group 10	Comparative Evaluation of LLM Metrics	Comparing EM, Accuracy, F1, BLEU, ROUGE, METEOR, BERTScore, WER, and CER to understand their strengths and limitations.

3. Requirements for Each Group

Each group must complete the following tasks:

1. **Introduction**

   Explain the assigned evaluation method and its importance in evaluating LLM performance.

2. **Theory**

   Explain the theoretical concepts, mathematical formulas, and underlying principles.

3. **Practical Examples**

   Provide at least two practical examples to demonstrate how the evaluation method works.

4. **Python Implementation**

   Implement the assigned evaluation method using Python. Students may use appropriate Python libraries.

5. **Experiment**

   Evaluate outputs from at least two LLMs using the assigned method, where applicable.

6. **Advantages and Limitations**

   Discuss the strengths, weaknesses, and limitations of the evaluation method.

7. **Conclusion**

   Explain when, why, and in which applications the evaluation method should be used.

4. Expected Deliverables

Each group must submit the following materials:

- Research report in PDF format.
- Python source code.
- Experimental results and comparison tables.
- Presentation slides.
- Live demonstration of the evaluation method.

5. Presentation and Discussion

Each group must present its research topic and demonstrate the implementation to the class.

Students should be prepared to:

- Explain the mathematical concepts and formulas.
- Demonstrate the Python implementation.
- Interpret the experimental results.
- Answer questions from the lecturer and classmates.
- Discuss the advantages and limitations of their assigned evaluation method.

6. Final Objective

By completing this assignment, students should be able to:

- Understand different evaluation methods used for Large Language Models.
- Explain and apply mathematical evaluation metrics.
- Implement evaluation methods using Python.
- Compare and analyze the performance of different LLMs.
- Select appropriate evaluation metrics for different NLP tasks.
- Identify the strengths and limitations of automatic evaluation methods.

Final Objective

Understand how to evaluate, measure, compare, and analyze the performance of Large Language Models using different evaluation metrics.